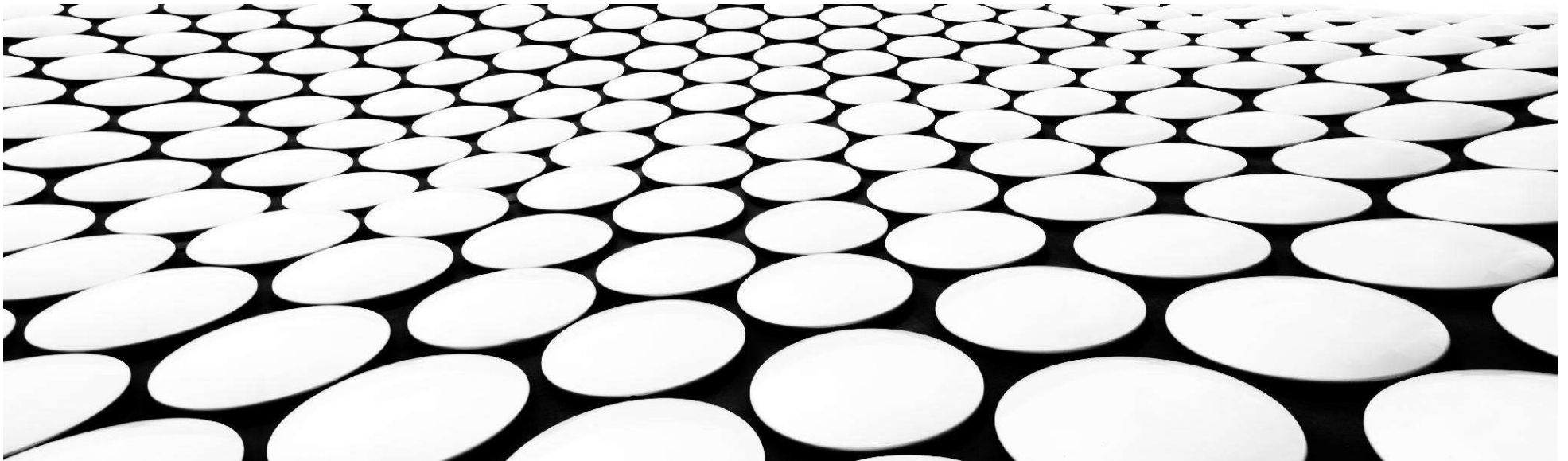

REDUCING ESTIMATION ERROR IN PORTFOLIO OPTIMIZATION

HOW DENOISING, DETONING, AND HRP IMPROVED A 10-ASSET PORTFOLIO



WHAT IS ESTIMATION ERROR?

Estimation error is the difference between the true but unknown portfolio parameters and the values estimated from historical data.

$$\hat{\Sigma} = \Sigma + \epsilon$$

Where $\hat{\Sigma}$ is the estimated covariance matrix,
 Σ is the true covariance matrix, and
 ϵ is estimation noise.

Small errors in covariance and correlation estimates can lead to unstable portfolio weights, especially when optimization methods rely heavily on historical data.

EVIDENCE FROM OUR PORTFOLIO

The constrained Kelly portfolio was concentrated in a few assets

Asset	Constrained Kelly Weight
TSLA	20.00%
GS	20.00%
META	20.00%
WMT	19.56%
GOOG	13.40%
BAC	7.04%
LLY, MRK, AAPL, XOM	0.00%

This means the model excluded four assets completely. That is not automatically wrong, but it shows sensitivity to the estimated return and covariance structure.

HOW THE METHODS ADDRESSED THE PROBLEM

- Denoising reduced random noise in the correlation matrix.
- Detoning removed the dominant market factor so that the model could focus more on diversifiable relationships.
- HRP allocated capital through clusters instead of relying on direct covariance matrix inversion.



This made the final portfolio more diversified and less sensitive to noisy estimates.

RESULTS AND CONCLUSION

Strategy	Return	Sharpe	Max Drawdown	95% cVaR	Variance
Constrained Kelly	7.46%	1.65	-6.86%	-2.45%	0.000135
Standard HRP	9.80%	2.48	-3.94%	-1.99%	0.000096
ML HRP	12.42%	3.50	-3.53%	-1.64%	0.000074

The ML-enhanced portfolio performed better out of sample. The improvement was not only in return but also in risk control. This supports the idea that reducing estimation error can improve portfolio construction.